Four Types of Detection (Traffic)

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- 1. Video Detection
- 2. Real-time Detection
  - 2.1 USB Camera

Four types of detection is a built-in example demonstration of DeepStream: detecting and counting people, vehicles, road signs, and bicycles in the video.

```
This tutorial only demonstrates the basics; other applications require user
development.
```

1. Video Detection

Enter the folder containing the example:

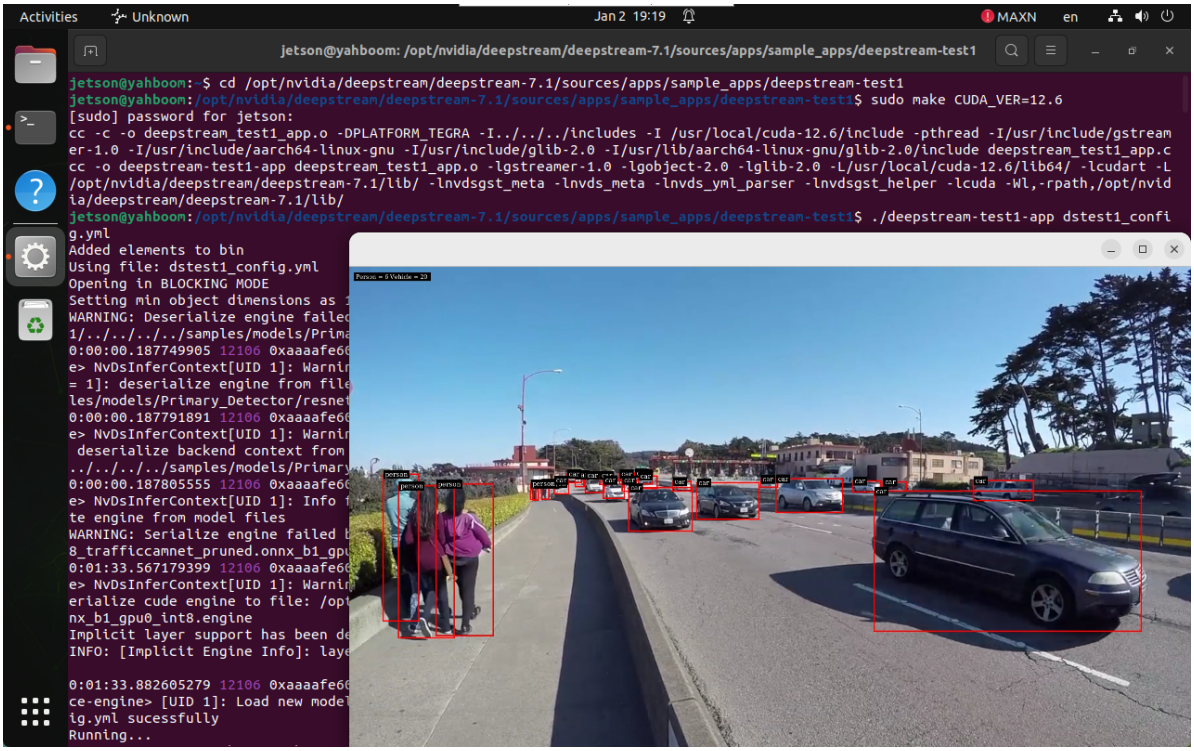
```
cd /opt/nvidia/deepstream/deepstream-7.1/sources/apps/sample_apps/deepstream-
test1
```

Compile the source code: Refer to the README file in the same folder.

```
sudo make CUDA_VER=12.6 # Compiled from the factory image, no user compilation
required.
```

Run the example: Refer to dstest1_config.yml in the same folder.

```
./deepstream-test1-app dstest1_config.yml
or
./deepstream-test1-app ../../../../samples/streams/sample_720p.h264
```



Configuration file: dstest1_config.yml

```
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################################################################################

source:
  location: ../../../../samples/streams/sample_720p.h264

streammux:
  batch-size: 1
  batched-push-timeout: 40000
  width: 1920
  height: 1080

# Inference using nvinfer:
primary-gie:
  plugin-type: 0
  config-file-path: dstest1_pgie_config.yml

## Inference using nvinferserver:
#primary-gie:
#  plugin-type: 1
#  config-file-path: dstest1_pgie_nvinferserver_config.txt
```
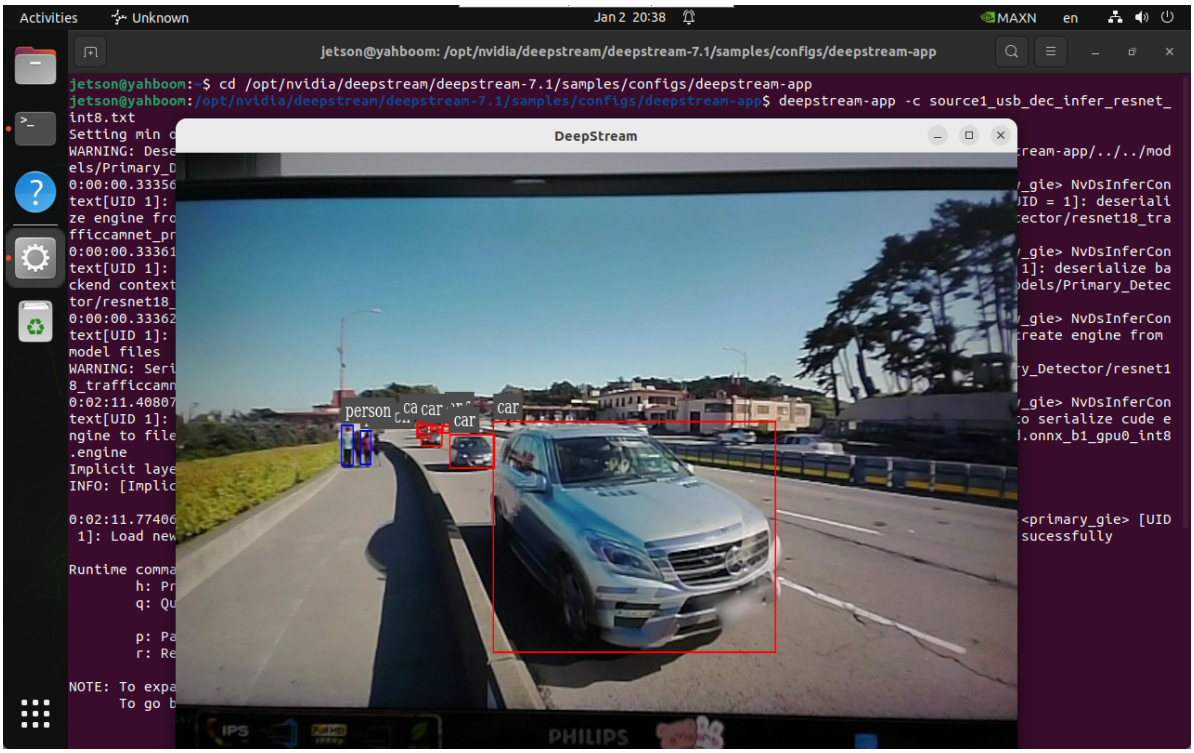

2. Real-time Detection

If there are multiple cameras, modify the .txt configuration file yourself: The following examples show the effect when the motherboard is connected to only one camera.

2.1 USB Camera

```
cd /opt/nvidia/deepstream/deepstream-7.1/samples/configs/deepstream-app
```

```
deepstream-app -c source1_usb_dec_infer_resnet_int8.txt
```